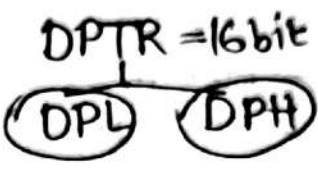


Mazidi

Addressing Modes (ଆବୃତ୍ତି ପଦ୍ଧତି)

- ① Register
- ② Direct
- ③ Register Indirect
- ④ Immediate (Constant PTR)
- ⑤ Indexed



Data Pointer Register
EQU
↳ Equate

We can set immediate values to ports

MOV P1, #55H

A Accumulator

MOV R1, R2 X

MOV A, #20H ✓
MOV R1, #20H
MOV A, R1
MOV R2, A

MOV DPTR, A X

↑ 16 bit ↑ 8 bit
incompatible register sizes

Direct Addressing Mode

MEMORY	VALUE
1000	5
1002	10
1004	15

MOV A, 4
MOV A, R4

they are the same

MOV A, 1000H

Memory Address

No hash sign

* Register banks (Bank 0)

R0	0
R1	1
R2	2
R3	3
R4	4

SFR

Special Function Register

ACC *	→	0E0 H
B *	→	0F0 H
P0	→	80 H
P1	→	90 H
P2	→	0A0 H
P3	→	0B0 H

Example 5.1 {P166}

(a) MOV A, #55
 MOV P1, A
 MOV P2, A

(b) MOV A, #55
 MOV 90H, A
 MOV A0H, A

PUSH(05H) ← Register
 PUSH(A0H) ← Address

Register Indirect

Only R0 & R1 are used

@R0
 @R1

pointer

MOV A, @R0
MOV @R1, B

R0	1000	5
R1	1002	10

Example 5.3

Memory	Value
40	55
41	55

(a) MOV A, #55H
MOV 40H, A
MOV 41H, A अक्षर
loop impossible
✓ in efficient

(b) MOV A, #55
MOV R0, #40H
MOV @R0, A
INC R0
MOV @R0, A

(c) MOV A, #55H
MOV R0, #40H
MOV R2, #02

AGAIN: MOV @R0, A
INC R0
DJNZ R2, AGAIN

Advantage?

- ✓ Dynamic data
- ✓ Indirect register & loop
- ✗ Direct & no loop

Exp 5.1

Write a program to clear 16 RAM locations from 60H

CLR A

MOV RI, #60H

MOV R7, #16

AGAIN MOV @RI, A

INC RI

DJNZ R7, AGAIN

R1, R2

Exp 5.5

10 bytes of data from 35H to 60H

MOV R0, #35H

MOV R7, #10

AGAIN: MOV @R0, R7

INC R0

DJNZ R7, AGAIN

~~MOV R0, #35H~~

~~MOV R7, #10H~~

~~MOV A, R0~~

~~AGAIN: MOV @R1, A~~

✓ MOV R0, #35H

✓ MOV R1, #60H

✓ MOV R7, #10H

✓ MOV A, @R0

✓ AGAIN: MOV @R1, A

✓ INC R0

✓ INC R1

✓ ~~DEC R7~~

✓ DJNZ R7, AGAIN

Exp 5.5

MOV R0, #35

MOV R1, #60

MOV R7, #10

BACK: MOV A, @R0

MOV @R1, A

INC R0

INC R1

DJNZ R7, BACK

Index Addressing Mode

✓ Look up Table

MOV A, @A+DPTR

↓
0 + 200
1 + 200
... + 200
n + 200

MOV A → MOV "C" (Code)

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Example 5.7

Example 5.8

Example 5.10

1111 1111
F F

DONE

// page 175 //

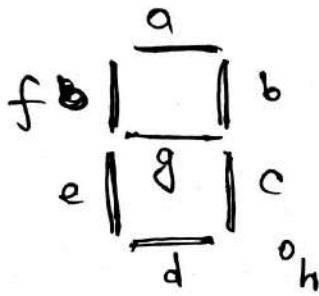
Pin 40 → VCC

Pin 20 → GND

Seven Segment

Anode

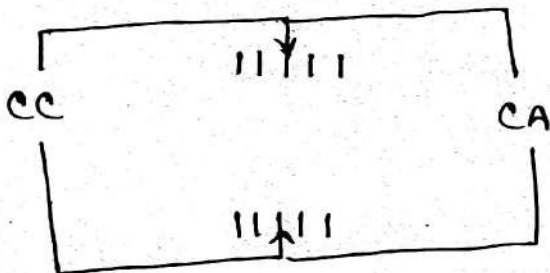
Cathode



0-9



dp → decimal pointer



CC = Common Cathode

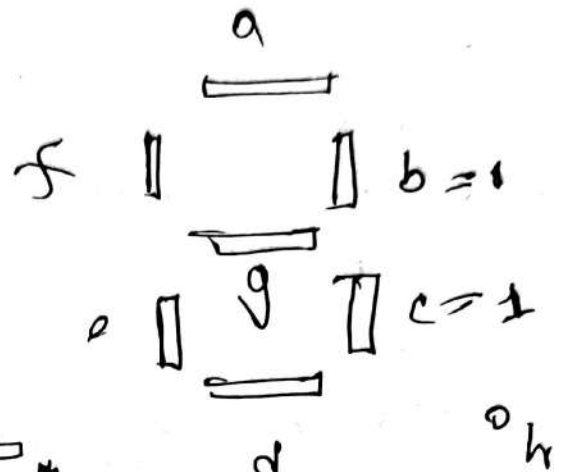
CA = Common Anode

g f a b

e d c dp

h	g	f	e	d	c	b	a
0	0	1	1	1	1	1	1

- 0 → 3FH
- 1 → 06H
- 2 → 5BH
- 3 → 4FH
- 4 → 66H
- 5 → 6DH
- 6 → 7DH
- 7 → 07H
- 8 → 7FH
- 9 → 67H



h	g	f	e	d	c	b	a
3FH	0	0	0	0	0	1	1

ORG 00H

MOV P2, #00H

MOV DPTR, #300H

MOV R0, #10D

SEVEN:

MOV A, #00H

MOVC A, @A+DPTR

MOV P2, A

LCALL DELAY

INC DPTR

DJNZ, R0, SEVEN

DELAY

MOV R2, #10

RAI:

MOV R3, #255